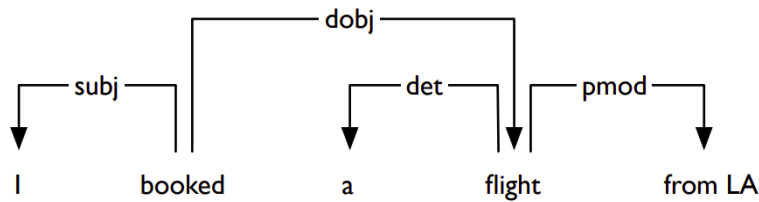


Q1. Draw the CKY table for the sentence - “A student likes writing tests” based on the grammar given below

Parent	Child / Children
S	NP VP
VP	VBG NNS
VP	VBZ VP
VP	VBZ NP
NP	JJ NNS
DT	“A”
NN	“student”
VBZ	“likes”
VBG	“writing”
JJ	“writing”
NNS	“tests”

0	A	1 student	2 likes	3 writing	4 tests	5
	DT χ_{01}	NP χ_{02}	— χ_{03}	— χ_{04}	$S_1 = NP VP_1$ $S_2 = NP VP_2$ χ_{05}	
		NN χ_{12}	— χ_{13}	— χ_{14}	— χ_{15}	
			VBZ χ_{23}	— χ_{24}	$VP_1 = VBZ VP$ $VP_2 = VBZ NP$ χ_{25}	
				VBG JJ χ_{34}	VP NP χ_{35}	
					NNS χ_{45}	

2. Given the following dependency graph please fill the table with the appropriate transitions



Transitions	Stack	Buffer	Arc

3. Suppose you typed ‘hoper’ by mistake instead of ‘hotel’. Explain how the symmetric delete spelling correction would work to provide the correct candidate ‘hotel’ but not words like ‘horse’, ‘chore’ etc.

4. Consider the following paragraph:

“Marie was born in 1867 in Warsaw, Poland, where her father was a professor of physics. At an early age, she displayed a brilliant mind and a blithe personality. Her great exuberance for learning prompted her to continue with her studies after high school. She became disgruntled, however, when she learned that the university in Warsaw was closed to women. Determined to receive a higher education, she defiantly left Poland and in 1891 entered the Sorbonne, a French university, where she earned her master’s degree and doctorate in physics.”

Create a bi-gram language model from the paragraph. Detect whether there is any real-word or non-word error in the following phrase - “She learned hert”. Assume the vocabulary to be only the words contained in the paragraph.

5. Suppose, you got a trained word2vec model where the following word vectors are given: drive - [0.3, 0.4], driver [0.8, 0.7], swim [0.4, 0.1]. From this can you calculate the word vector for the word “swimmer”?